# Lecture 1: Mathematical Foundations for Lattice-Based Cryptography

Groups, Rings, Fields, Modular Arithmetic,
Vector Spaces, Basis, and Lattices

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Lecture Notes — Session 1 (approx. 2 hours)

**Purpose of this lecture.** Every technical term used later in the thesis proposal — “lattice,” “modulus,” “ring $\mathbb{Z}_q[x]/(x^n+1)$” — rests on a small set of algebraic building blocks. This lecture builds those blocks, one at a time, from the ground up: *Group* $\rightarrow$ *Ring* $\rightarrow$ *Field* $\rightarrow$ *Modular Arithmetic* $\rightarrow$ *Vector Space* $\rightarrow$ *Basis* $\rightarrow$ *Lattice*. Nothing here requires prior abstract algebra; every definition is followed immediately by concrete, worked examples.

## Contents



## 1 Why Start With Algebra?

Lattice-based cryptography is, at its core, about doing arithmetic on *vectors* whose coordinates live in a *finite, wraparound number system*, arranged according to strict algebraic rules. Before we can even state what a lattice is, we need precise answers to questions like:

- What does it mean for a set of numbers to have a well-behaved notion of “add” and “multiply”? (**Groups, Rings, Fields**)
- What happens when arithmetic “wraps around,” like a clock? (**Modulus**)
- What does it mean to build every point in a space out of a small number of building-block directions? (**Vector spaces, Basis**)
- What happens when we only allow *whole-number* combinations of those building blocks, instead of

any real-number combination? (**Lattice**)

This lecture answers each of these in order. By the end, the definition of a lattice (Section 7) should feel like the natural, inevitable next step, not an isolated new idea.

## 2 Sets and Binary Operations (Quick Refresher)

A **set** is just a collection of objects (numbers, vectors, symbols — anything). A **binary operation** on a set $S$ is a rule that takes two elements of $S$ and combines them into a third element, e.g., ordinary addition $+$ takes two numbers and produces their sum. We write $a * b$ for a general binary operation.

An operation is **closed** on $S$ if combining two elements of $S$ always gives another element of $S$ (never something outside $S$). This sounds obvious but fails more often than you'd expect — see Example 1 below.

**Example: Closure Can Fail**

Let $S = \{1, 2, 3, 4, 5\}$ (just these five numbers) and let $*$ be ordinary addition. Then $2 * 4 = 6$, but $6 \notin S$. So ordinary addition is *not* closed on $S$. This is exactly the kind of problem modular arithmetic (Section 5) is built to fix.

## 3 Groups

**Definition: Group**

A set $G$ together with a binary operation $*$ is called a **group**, written $(G, *)$, if all four of the following hold:
(G1) **Closure:** for all $a, b \in G$, $a * b \in G$.
(G2) **Associativity:** for all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
(G3) **Identity:** there exists an element $e \in G$ such that $a * e = e * a = a$ for all $a \in G$.
(G4) **Inverses:** for every $a \in G$, there exists $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.
If, in addition, $a * b = b * a$ for all $a, b \in G$ (the order doesn't matter), the group is called **abelian** (commutative).

**Example: Groups You Already Know**

- $(\mathbb{Z}, +)$ — the integers under addition. Identity is $0$; the inverse of $a$ is $-a$. This is an abelian group.
- $(\mathbb{R} \setminus \{0\}, \times)$ — nonzero real numbers under multiplication. Identity is $1$; the inverse of $a$ is $1/a$. Also abelian. (We had to remove $0$ because $0$ has no multiplicative inverse.)
- $(\mathbb{Z}, \times)$ — integers under multiplication — is **not** a group: $2$ has no inverse in $\mathbb{Z}$ (we'd need $1/2$, which isn't an integer). This fails axiom (G4).

**Remark: Why Cryptography Cares About Groups**

Many classical cryptographic schemes (Diffie–Hellman, ElGamal, elliptic-curve cryptography) are built directly on top of group structure — the security relies on certain problems (like "given $g$ and $g^a$, find $a$") being hard inside a specific group. Lattice-based cryptography instead builds on richer structures (rings and modules, coming next), but the group is always the first layer underneath.

**Exercise: Try This Before Next Lecture**

Is $(\mathbb{Z}_5 \setminus \{0\}, \times \bmod 5) = (\{1, 2, 3, 4\}, \times \bmod 5)$ a group? Check all four axioms directly by building the multiplication table. (You'll need Section 5 for what "mod5" means if it's new to you — come back to this after reading that section.)

## 4 Rings

A ring is what you get when you ask a set to support *two* operations — usually called "+" and "×" — that interact the way ordinary addition and multiplication do.

**Definition: Ring**

A set $R$ with two binary operations $+$ and $\times$ is a **ring** $(R, +, \times)$ if:
(R1) $(R, +)$ is an abelian group (with identity element usually written $0$).
(R2) $\times$ is associative: $(a \times b) \times c = a \times (b \times c)$.
(R3) The distributive laws hold: $a \times (b + c) = (a \times b) + (a \times c)$ and $(a + b) \times c = (a \times c) + (b \times c)$.
Note: a ring does *not* need multiplicative inverses, and multiplication need not be commutative in general (though in every ring we use in this thesis, it will be — these are called **commutative rings**). If there is an element $1 \in R$ with $a \times 1 = 1 \times a = a$ for all $a$, we say $R$ is a **ring with unity**.

**Example: Rings You Already Know**

- $(\mathbb{Z}, +, \times)$ — the integers form a commutative ring with unity $1$. Notice $2$ still has no multiplicative inverse, and that's fine — rings don't require one.
- **Polynomial rings**, e.g. $\mathbb{Z}[x]$: the set of all polynomials with integer coefficients (like $3x^2 - x + 7$), with ordinary polynomial addition and multiplication. This is a ring for exactly the same reason $\mathbb{Z}$ is.

**Looking Ahead: The Ring That Kyber and Dilithium Actually Use**

Both Kyber and Dilithium do their arithmetic inside a specific ring of the form

$$R_q = \mathbb{Z}_q[x]/(x^n + 1),$$

read as: "polynomials with coefficients taken modulo $q$ (Section 5), where we also reduce powers of $x$ using the rule $x^n \equiv -1$." This looks intimidating right now, but it is built from exactly the pieces in this lecture: a polynomial ring (this section), with coefficients from $\mathbb{Z}_q$ (Section 5), further "folded" by a fixed rule. We will unpack this fully in a later lecture — for now, just notice that "ring" is not an abstract curiosity; it's the literal object the cryptography is built on.

## 5 Fields

**Definition: Field**

A **field** is a commutative ring with unity $F$ in which *every nonzero element has a multiplicative inverse* — that is, $(F \setminus \{0\}, \times)$ is also an abelian group. Informally: a field is a set where you can add, subtract, multiply, and divide (by anything except 0) and always stay inside the set.

**Example: Fields and Non-Fields**

- $\mathbb{Q}$ (rationals), $\mathbb{R}$ (reals), $\mathbb{C}$ (complex numbers) are all fields — you can divide by any nonzero element and stay inside the set.
- $\mathbb{Z}$ is **not** a field: $2 \in \mathbb{Z}$ has no multiplicative inverse in $\mathbb{Z}$ (again, $1/2 \notin \mathbb{Z}$). $\mathbb{Z}$ is a ring, but not a field.
- $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$ under addition and multiplication mod 5 *is* a field — every nonzero element has an inverse (you checked this, or will check this, in Exercise 4.1). Fields built this way, from $\mathbb{Z}_p$ with $p$ prime, are called **finite fields** or **Galois fields**, written $\mathbb{F}_p$ or $GF(p)$.

**Remark: The One-Sentence Summary So Far**

Group $\subset$ Ring $\subset$ Field, in the sense that a field is a ring with extra structure (division), and a ring is built from a group with extra structure (a second operation). Each step adds a capability: groups let you undo one operation (inverses); rings add a second, distributive operation; fields guarantee *both* operations are fully invertible (except division by 0).

## 6 The Modulus: Arithmetic That Wraps Around

**Definition: Congruence Modulo $n$**

For a fixed positive integer $n$ (the **modulus**), two integers $a$ and $b$ are **congruent modulo** $n$, written

$$a \equiv b \pmod{n},$$

if $n$ divides $(a - b)$ evenly — equivalently, $a$ and $b$ leave the same remainder when divided by $n$. The set of possible remainders, $\{0, 1, 2, \ldots, n - 1\}$, together with addition and multiplication performed "with wraparound," is written $\mathbb{Z}_n$ (or $\mathbb{Z}/n\mathbb{Z}$).

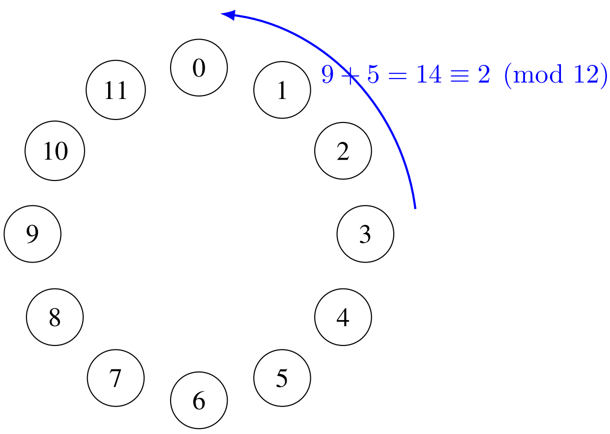


Figure 1: The clock analogy for modular arithmetic: $\mathbb{Z}_{12}$ works exactly like a 12-hour clock face. 9 o'clock plus 5 hours wraps around to 2 o'clock, i.e., $9 + 5 = 14 \equiv 2 \pmod{12}$.

**Example: Computing Modulo**

$17 \bmod 5$: divide 17 by 5 to get 3 remainder 2, so $17 \equiv 2 \pmod{5}$. Similarly, $23 \equiv 3 \pmod{10}$, and $-1 \equiv n - 1 \pmod{n}$ (e.g., $-1 \equiv 4 \pmod{5}$) — negative numbers wrap around too.

**Remark: When Is $\mathbb{Z}_n$ a Ring? When Is It a Field?**

$\mathbb{Z}_n$ is *always* a commutative ring with unity, for any $n \geq 2$. But it is a *field* only when $n = p$ is **prime**. If $n$ is composite (e.g., $n = 6$), some nonzero elements fail to have multiplicative inverses — e.g., in $\mathbb{Z}_6$, there is no $x$ with $2x \equiv 1 \pmod 6$, since $2$ shares a common factor with $6$. This is exactly why cryptographic schemes that need a field (rather than just a ring) are built with a prime modulus $q$.

**Looking Ahead: Why Cryptography Loves the Modulus**

Modular arithmetic gives us two things cryptography needs badly: (1) a **finite** number system (so keys and computations have a fixed, bounded size, no matter how much you compute), and (2) built-in **one-wayness** in places — e.g., given only $a \bmod n$, you cannot tell which of infinitely many integers $a$ actually was. The Learning-With-Errors (LWE) problem from the proposal document hides its secret inside equations that live in exactly this kind of finite, modular number system (there, the modulus is usually called $q$).

**Exercise: Practice**

(a) Compute $23 \times 15 \bmod 7$. (b) List all elements of $\mathbb{Z}_8$ that *do not* have a multiplicative inverse, and explain why (hint: compare each element to $8$). (c) Confirm $\mathbb{Z}_7$ is a field by checking that 3 has a multiplicative inverse mod 7 (find $x$ such that $3x \equiv 1 \pmod 7$).

# 7 Vector Spaces, Basis, and Dimension

**Definition: Vector Space**

A **vector space** over a field $F$ is a set $V$ (whose elements are called **vectors**) equipped with vector addition and scalar multiplication (multiplying a vector by an element of $F$), satisfying the "obvious" rules you'd expect: addition makes $(V, +)$ an abelian group, and scalar multiplication distributes sensibly over both vector addition and field addition. The most important example for us: $\mathbb{R}^n$, the set of all $n$-tuples of real numbers, is a vector space over $\mathbb{R}$.

**Definition: Linear Combination, Span, Linear Independence**

Given vectors $v_1, \ldots, v_k \in V$ and scalars $c_1, \ldots, c_k \in F$, the vector $c_1v_1 + c_2v_2 + \cdots + c_kv_k$ is called a **linear combination** of $v_1, \ldots, v_k$. The set of *all* linear combinations of a set of vectors is called their **span**. The vectors $v_1, \ldots, v_k$ are **linearly independent** if the only way to write $0 = c_1v_1 + \cdots + c_kv_k$ is with every $c_i = 0$ — informally, no vector in the set is "redundant" (expressible using the others).

**Definition: Basis and Dimension**

A set of vectors $B = \{b_1, \ldots, b_k\}$ is a **basis** of a vector space $V$ if (1) $B$ is linearly independent, and (2) $B$ **spans** $V$ (every vector in $V$ can be written as a linear combination of $B$). Every basis of a given vector space has the same number of elements $k$, called the **dimension** of $V$.

**Example: Basis of $\mathbb{R}^2$**

The vectors $e_1 = (1,0)$ and $e_2 = (0,1)$ form the "standard" basis of $\mathbb{R}^2$: any vector $(a,b)$ is exactly $a \cdot e_1 + b \cdot e_2$. But this is far from the *only* basis — $\{(1,1),(1,-1)\}$ is also a valid basis of $\mathbb{R}^2$ (check: they're linearly independent, and any $(a,b)$ can be written as a combination of them). A vector space almost always has infinitely many possible bases; some are more convenient to work with than others. **Keep this fact in mind — it becomes very important in Section 7.**

## 8 From Vector Spaces to Lattices

Everything so far has allowed *any* scalar from the field $F$ (e.g., any real number) when forming linear combinations. A lattice is what happens when we impose one dramatic restriction: **only integer combinations are allowed.**

**Definition: Lattice**

Given $n$ linearly independent vectors $b_1, \ldots, b_n \in \mathbb{R}^n$ (called a **basis** of the lattice), the **lattice** they generate is the set of *all integer* linear combinations:

$$\mathcal{L}(b_1, \ldots, b_n) = \left\{ \sum_{i=1}^{n} a_i b_i \;\middle|\; a_i \in \mathbb{Z} \right\}.$$

Unlike a vector space (which is a continuous, solid region filling all of $\mathbb{R}^n$), a lattice is a **discrete** set of isolated points — it looks like an infinite, evenly-repeating grid, as in Figure 2. Formally, a lattice is a discrete additive subgroup of $\mathbb{R}^n$: it is itself a group under vector addition (check: it's closed, has an identity $\vec{0}$, every point has an inverse, and addition is associative), but its elements do not fill up space continuously.

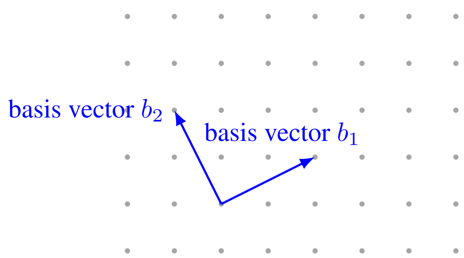


Figure 2: A 2-dimensional lattice generated by a basis $\{b_1, b_2\}$: every grid dot is reachable by some *integer* combination $a_1 b_1 + a_2 b_2$ (with $a_1, a_2 \in \mathbb{Z}$), starting from any chosen origin dot.

**Remark: Same Lattice, Different Basis**

Just as with vector spaces (Example 7), a single lattice has *many* different valid bases. Two bases generate the exact same set of grid points if and only if one can be converted to the other using an integer matrix with determinant $\pm 1$ (a **unimodular** transformation) — we won't need the details of this fact yet, but the consequence is critical: some bases consist of short, nearly-perpendicular vectors (a "good" basis, easy to compute with), while other, equally valid bases for the *same* lattice consist of long, nearly-parallel vectors (a "bad" basis, hard to compute with). **This gap between good and bad bases for the same lattice is the single most important idea in lattice-based cryptography**: the secret key is typically a good (short) basis, while the public key describes the same lattice using a bad (long, scrambled-looking) basis. Recovering the good basis from the bad one is believed to be extremely hard once the dimension $n$ is large enough.

**Example: A Tiny Lattice by Hand**

Let $b_1 = (2,0)$ and $b_2 = (0,3)$ in $\mathbb{R}^2$. The lattice $\mathcal{L}(b_1, b_2)$ consists of every point $(2a_1, 3a_2)$ for integers $a_1, a_2$ — so it includes $(0,0)$, $(2,0)$, $(-2,0)$, $(0,3)$, $(2,3)$, $(4,-3)$, and so on, but *not* $(1,1)$ or $(1,0)$, since those cannot be written as $(2a_1, 3a_2)$ for integer $a_1, a_2$.

**Looking Ahead: Where This Leads Next Lecture**

With Groups → Rings → Fields → Modular arithmetic → Vector spaces → Basis → Lattice now in place, the next lecture can finally state the **Learning With Errors (LWE)** problem precisely: a secret vector, hidden inside noisy linear equations over $\mathbb{Z}_q$ (a ring, Section 5), which is exactly the kind of "bad basis" hardness described above, dressed up in a specific, standardized cryptographic form. We will also unpack the ring $R_q = \mathbb{Z}_q[x]/(x^n + 1)$ flagged in Section 4, since **module lattices** (lattices whose coordinates come from a ring like $R_q$ instead of plain integers) are exactly what Kyber and Dilithium use in practice.

## 9 Summary Table

| Structure | Requires | Example |
|---|---|---|
| Group | 1 operation, inverses exist | $(\mathbb{Z}, +)$ |
| Ring | 2 operations, no division needed | $(\mathbb{Z}, +, \times)$, $\mathbb{Z}[x]$ |
| Field | Ring + division by nonzero elements | $\mathbb{Q}, \mathbb{R}, \mathbb{Z}_p$ ($p$ prime) |
| Modulus | Wraparound arithmetic | $\mathbb{Z}_n = \{0, \dots, n-1\}$ |
| Vector space | Field + vectors + scalar mult. | $\mathbb{R}^n$ |
| Basis | Independent, spanning vector set | $\{(1,0), (0,1)\}$ in $\mathbb{R}^2$ |
| Lattice | Vector space, but *integer* combos only | Grid of points in $\mathbb{R}^n$ |

## 10 Full List of Exercises (Assign as Homework Before Lecture 2)

**Exercise: Homework Set**

1. Verify $(\{1,2,3,4\}, \times \bmod 5)$ is a group by writing out its full multiplication table (Exercise 4.1 above).
2. Compute $23 \times 15 \bmod 7$; list the non-invertible elements of $\mathbb{Z}_8$; verify $\mathbb{Z}_7$ is a field by finding the inverse of 3 (Exercise 6.1 parts a–c above).
3. Is $\mathbb{Z}_9$ a field? Why or why not? (Hint: think about what 9 is *not*.)
4. Show that $\{(1,1), (1,-1)\}$ is a valid basis of $\mathbb{R}^2$ by (a) checking linear independence and (b) writing the vector $(4,2)$ as a linear combination of them.
5. For the lattice generated by $b_1 = (1,2)$ and $b_2 = (3,1)$: is the point $(7,5)$ in the lattice? (Find integers $a_1, a_2$ with $a_1 b_1 + a_2 b_2 = (7,5)$, or show none exist.)
6. In your own words (3–4 sentences), explain why a "good basis" and a "bad basis" for the same lattice can make the same mathematical object feel completely different to work with computationally.